CSE321 Introduction to Algorithm Design HW1

Some Reminders

Informally

$$\begin{array}{ccc} O & \approx & \leq \\ \Omega & \approx & \geq \\ \Theta & \approx & = \\ o & \approx & < \\ \omega & \approx & > \end{array}$$

Formally

O-notation

$O(g(n)) = \{f(n) :$ there exist positive constants c and n_0 such that $0 \leq f(n) \leq cg(n)$ for all $n \geq n_0\}$.

Ω-notation

$\Omega(g(n)) = \{f(n) :$ there exist positive constants c and n_0 such that $0 \leq cg(n) \leq f(n)$ for all $n \geq n_0\}$.

Θ-notation

$\Theta(g(n)) = \{f(n) :$ there exist positive constants c_1, c_2, and n_0 such that $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0\}$.

o-notation

$o(g(n)) = \{f(n) :$ for all constants $c > 0$, there exists a constant $n_0 > 0$ such that $0 \leq f(n) < cg(n)$ for all $n \geq n_0\}$.

ω-notation

$\omega(g(n)) = \{f(n) :$ for all constants $c > 0$, there exists a constant $n_0 > 0$ such that $0 \leq cg(n) < f(n)$ for all $n \geq n_0\}$.

1-)Show the following. For each question state specific constants (eg. c_1, c_2, n_0) . Remember that just stating constants is not enough. You should **prove** that for any x value the expression is **always true** when the stated constants used . (so showing that the expression is true just for some n values is not enough and such solutions will not be graded.)

a) $4n^3 - 5n^2 + 7n + 9 = \theta(n^3)$

b) $3n^2 + 4n - 5 = O(n^2 - 3n + 1)$

c) $3n^2 + 5n(lgn)^3 + 9n = O(n^2)$

2-)Compare the orders of growth of (use limit)

a) $(sin\pi y)^2$ and $(y^2 + 1)$

b) $(ln x)^2$ and x

c) $arctan(x)$ and $1/\arcsin(1/x)$

3-). Give the running time of the code snippet below depending on n.

```
for (i=2*n; i>=1; i=i-1) :
      for (j=1; j<=i; j=j+1) :
                for (k=1; k<=j; k=k*3) :
                      print("merhaba")
```

4-)Compute the following sums .

a) $\sum_{m=1}^{\infty} \frac{1}{m^3}$

b) $\sum_{m=1}^{\infty} \frac{1}{m^2+1}$

5-) Design an algorithm that finds the smallest k elements in a list of n numbers in

a) θ(kn) time.
b) O(n) time at average case